

Maryam Ahmed Adel

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EDUCATION

Bachelor of Computers and Artificial Intelligence, Information Systems Department *Expected: Jun 2028*

Cairo University, Cairo | Third-Year Student | GPA: 3.17/4.0

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Statistics & Probability, Linear Algebra, Software Engineering, Structured Programming

SUMMARY

Information Systems student at Cairo University with a strong academic and practical focus on Artificial Intelligence. Experienced in developing and applying Machine Learning, Deep Learning, Computer Vision, and Generative AI solutions through internships, academic projects, and independent work. Committed to continuous learning and professional development, with a particular interest in building reliable, practical AI systems that address real-world challenges and deliver meaningful impact.

SKILLS

Generative AI / NLP: LLMs, RAG, LangChain, LangGraph, Hugging Face, Fine-tuning, Prompt Engineering

ML / DL Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn

Languages: Python, C++, Java, SQL

Tools & Deployment: FastAPI, Streamlit, Docker, Git, GitHub Actions, Linux/Unix

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn

Other: Computer Vision, Arduino, AI Automation, MLOps

EXPERIENCE

Software Intern — EAEA (Egyptian Atomic Energy Authority) *Cairo | Aug 2026 – Sep 2026*

- Explored the intersection of software, AI, and nuclear technology
- Gaining practical knowledge of embedded systems and their applications

AI Intern — MNKI *Cairo | Jun 2026 – Aug 2026*

- Worked in a real-world AI project focused on transforming NLP into CAD designs
- Exploring how Generative AI can support architectural designs

AI Intern — CIB (Commercial International Bank) *Cairo | Aug 2025 – Sep 2025*

- Focused on Generative AI applications for business use cases during an intensive corporate AI training program.
- Gained exposure to corporate-level data workflows and security standards for machine learning applications in financial services.

AI Intern — GTC *Cairo | Aug 2025 – Sep 2025*

- Built a face detection system by fine-tuning FaceNet and deployed it as an interactive Streamlit application.
- Refined data preprocessing pipelines for industry-specific machine learning challenges.

Technical Freelancer — ITIDA Gigs *Cairo | May 2025 – Jul 2025*

- Completed technical project tasks under the ITIDA initiative, meeting industry-standard deadlines and requirements.

PROJECTS

AI Zatoona [GitHub](#)

- Built the AI layer of a research-paper summarization system using LangGraph to orchestrate the pipeline, generating structured summaries (methods, experiments, limitations) with Hugging Face open-source models.
- Collaborated with a teammate to ship a full-stack product (FastAPI backend, React Native app, Postgres, Tailwind) with GitHub Actions CI/CD, deployed and live for public use.

- Designed a habit-forming "Reading Streaks" feature and Notion-integrated archiving so users build a searchable library of research they've covered.

AI Proactive Road Safety System (APRSS) [GitHub](#)

- Contributed the Streamlit monitoring dashboard for a team-built proactive collision-risk system (CARLA + SUMO co-simulation → YOLOv8 perception → XGBoost risk classifier → FastAPI → Docker → Azure).
- Designed a three-panel command-room layout with a real-time YOLOv8 perception overlay, image/video inspection modes, and a session summary with a risk-over-time timeline.

Fire & Smoke Detection System (Computer Vision)

- Collaborated with a technical team to build a computer vision hazard detection system.
- Performed data augmentation on Roboflow datasets and deployed the solution via Streamlit.

Flight Delay Prediction (Neural Network & MLOps) [GitHub](#)

- Designed and optimized a Neural Network model, achieving a 90% R^2 score.
- A system to predict the delay for the flights

Self-Driving Car Prototype | *Cairo University*

- Developed an autonomous vehicle prototype using Arduino and sensor integration for obstacle avoidance.
- Programmed hardware-software interfaces to translate sensor inputs into real-time mechanical movement.

CERTIFICATES

- Machine learning engineer training program — DEPI (Ministry of Communications and Information Technology)
- AI Diploma — Dotpy
- HCIA-AI
- Intro to Generative AI — AWS via Udacity
- Responsible Innovation and Trustworthy AI (Badge) — SAS
- RAG — Coursera
- Crew AI — Coursera
- Problem Solving Level 1 & 2 — Coach Academy
- Career Direction Program — EFE (Education For Employment)